

CPIT - 240 | Final Exam Fall 24

By: Z

Part 1: Mapping ER to Relational Schema (Chapter 9) [5 marks]

Q1. Given the following ER diagram **Customer → (1:N) → Order → (M:N) → Product** , Draw the relational schema

Essay Questions

Part 2: Functional Dependency and Normalization [9 Marks]

Q2. Define **normalization** and explain its **benefits** in database design.

Q3. What is **functional dependency**? Provide an example .

You are given a relation that is **not normalized** and are asked to check whether it violates the **1st, 2nd, and 3rd normal forms** (1NF, 2NF, and 3NF).

Q4:

StudentID	Name	Courses
101	John Smith	Math, Science
102	Jane Doe	English, History

1. **Provide the reason for the violations** and highlight the **problematic attributes**.
2. **Normalize** the relation & **Update the schema** and show the final normalized schema.

Q5:

OrderID	ProductID	ProductName	Quantity	OrderDate
1001	P01	Laptop	2	2024-12-01
1001	P02	Mouse	1	2024-12-01

1. **Provide the reason for the violations** and highlight the **problematic attributes**.
2. **Normalize** the relation & **Update the schema** and show the final normalized schema.

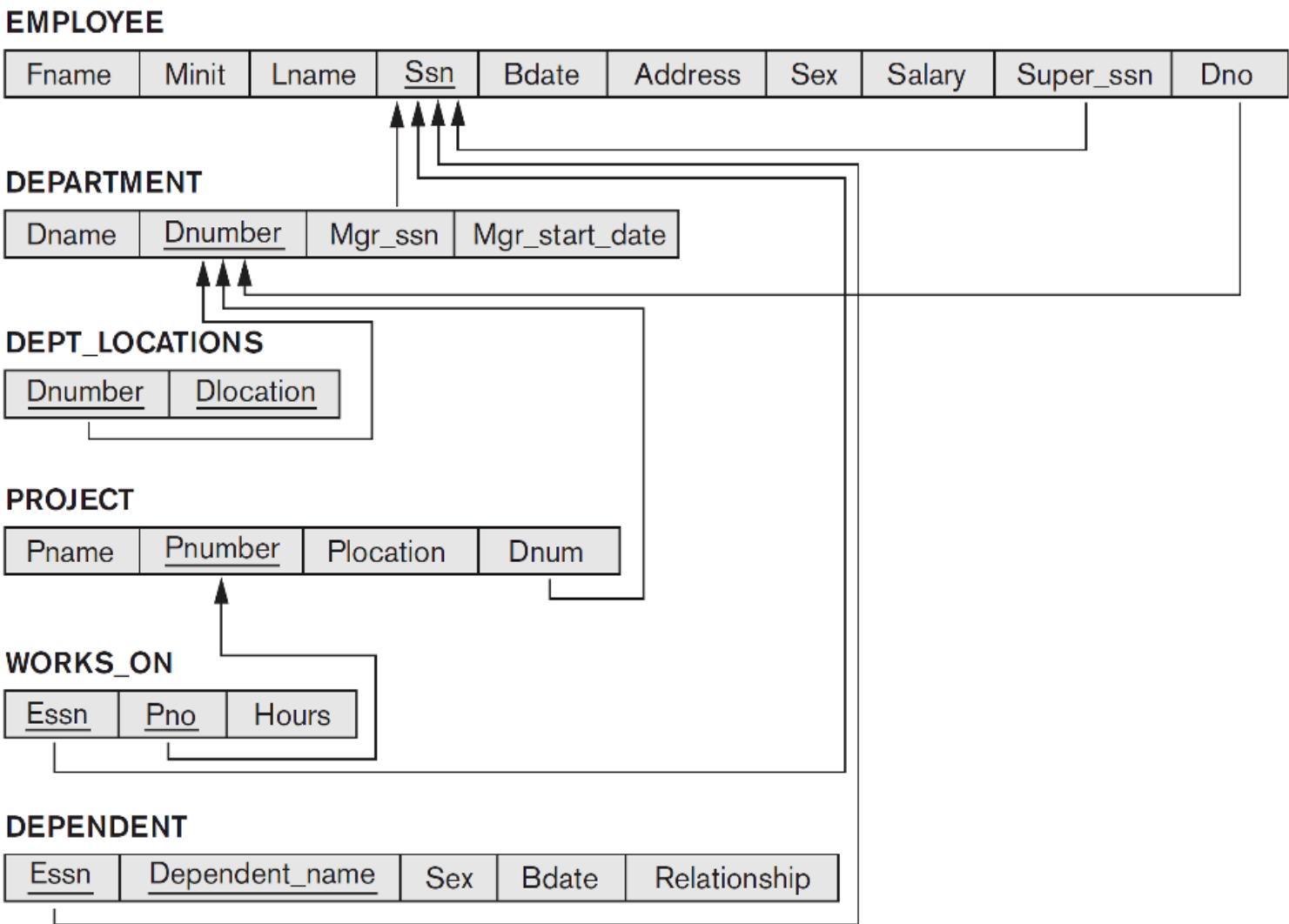
Q6:

EmployeeID	EmployeeName	DepartmentID	DepartmentName
1	Alice	D01	HR
2	Bob	D02	Finance

1. **Provide the reason for the violations** and highlight the **problematic attributes**.
2. **Normalize** the relation & **Update the schema** and show the final normalized schema.

Part 3: SQL (Chapters 6 and 7) [9 Marks]

Q7-15 Using the following company database Schema , write a **SQL query's** For the Following :



8 or 9 SQL Query Questions .

Part 4: Relational Algebra (Chapter 8)

اسئلة مشابهة للإختبار

Question Q16 -20	True/False
The intersection operation in relational algebra removes duplicates by default.	
The Cartesian product in relational algebra pairs every tuple in the first relation with every tuple in the second relation.	
A union operation in relational algebra requires both relations to have the same schema (same attributes and domains).	
The natural join operation in relational algebra eliminates duplicate columns automatically in the result.	

Question Q16 -20	True/False
The intersection operation can only be performed on relations with the same schema (same attributes and domains).	

21. Write a relational algebra expression to retrieve the following information The Number **of the department** , The **total number of employees** in the department. The **maximum salary** of employees in the department from EMPLOYEE Relation Group by Department .
22. Write a relational algebra expression to retrieve the first name, last name, and salary of employees who work in the "Research" department.

23. Given the following two tables [4 Marks] :

Customers:

Name	City
John	Cairo
Maria	Dubai
Ahmed	Cairo
Ali	Mansour

Suppliers:

Name	City
Zayed	Cairo
Yara	Dubai
Lina	Riyadh

Perform the following operations :

- a. Customers $\cup$ Suppliers
- b. Customers $\cap$ Suppliers
- c. Customers $-$ Suppliers
- d. Suppliers $-$ Customers

مهم :

ركزوا على الـ SQL & Relation Algebra
وتدربوا على امثله كثير ببسهل عليكم في الاختبار . .

بالتوفيق للجميع، ولا تنسونا من دعواتكم الطيبة .